

Saumik Dana

<https://saumikdana.github.io/>

Summary

US green card holder who builds and operates live ML- and LLM- driven systems end-to-end – research, deployment and infrastructure as one workflow. PhD from The University of Texas at Austin.

Experience

Quantitative Researcher, Zebra Capital Management LLC, Stamford CT Feb 2024 – Dec 2025
LLM-backed Trading Systems

- Deployed an intraday FX trading system that combined RSS news and price action into structured market narratives, generated trade decisions with *Llama-3.1-8B-Instant*.
- Closed the FX research-to-production loop with evaluation-gated prompt-optimization loop (DSPy/GEPA) using historical trading outcomes, promoting a new prompt when it outscored the incumbent.
- Deployed an intraday commodity trading workflow combining news, price action, and sovereign yield-curve differentials with *gpt-oss-20b* as a preprocessing filter.
- Engineered a mixture-of-experts pipeline using *gpt-oss-120b*, *DeepSeek-V3.1*, and *MiniMax-M2* for independent analysis, assigning each model a role based on its observed reasoning bias before consolidating into trade decisions.
- Implemented a *gpt-oss-120b*-driven intraday options strategy that translated calibrated volatility signals and market context directly into structured trading decisions.

ML-backed Trading Systems

- Deployed intraday options strategies using Bayesian anomaly-detection signals derived from stochastic-volatility model calibrations; generated *weekly returns of 10% on capital at risk*.
- Built and operated a live equity portfolio using Hidden Markov Models for market-regime detection, Jensen–Shannon divergence for rebalancing, and NSGA-II bi-objective optimization; generated *5% annualized alpha versus the benchmark in a multi-year backtest*.
- Built a research-to-production interday options platform spanning historical simulation, feature generation, model training, signal gating, and live trade execution across 18 equity and ETF underlyings.
- Fine-tuned Amazon Chronos-T5 with LoRA to generate rolling embeddings from technical indicators, then trained per-asset TabPFN classifiers to identify long-straddle opportunities.

LLM-backed Agents

- Built an MCP-compatible, tool-calling trade-signal mining agent that orchestrated hypothesis generation, data analysis, strategy construction, and backtesting; discovered strategies with *Sharpe ratios above 1.5*.
- Built a multimodal PDF market-research assistant that generated document-specific page captions, fine-tuned CLIP projection layers using contrastive InfoNCE training, and indexed page-image embeddings in Qdrant.
- Implemented similarity, score-threshold, and maximal-marginal-relevance retrieval with a two-pass Qwen vision-model workflow for grounded question answering and self-evaluated response generation.
- Delivered the PDF research system as a full-stack application using FastAPI, Next.js, persistent document workspaces, retrieval traces, and containerized Modal deployment.

Production Infrastructure

- Designed and deployed containerized serverless trading and research systems using AWS Lambda, ECR, S3, DynamoDB, EventBridge, CloudFormation, GitHub Actions, Docker, and Modal.
- Integrated Groq and Fireworks AI inference with HMM, Bayesian/Heston, Chronos, TabPFN, Qdrant, DSPy, and broker APIs within end-to-end production workflows.
- Engineered DST-aware market scheduling, persistent workflow and position state, model-artifact caching, API services, monitoring dashboards, execution recovery, and fail-closed error handling across multi-asset trading sessions.

Computational Engineer, VISIE Inc., Austin TX, Aug 2023 - Nov 2023

- Implemented TCP/UDP communication protocols to control robotic arm motion in surgical navigation platform
- Supported product packaging, deployment, live demonstrations *contributing to a successful \$8.2M Series A round*

Computational Lead, Sophelio, Austin TX, Aug 2022 - Mar 2023

- Adapted physics-informed modeling and sparse-regression PDE discovery to stock price prediction
- Built a swing-trading system using Bayesian TPE optimization to maximize CAGR

Postdoc, Los Alamos National Lab & University of Southern California, Jan 2019 - July 2022

- Replaced 3D fracture geometry with graph representation, running flow and transport as a reduced-order model
- Built sequential iterative multiphysics solver on grids tied together with computational-geometry projection layer

Education

PhD, Engineering Mechanics — University of Texas at Austin, Austin TX

MEng, Mechanical Engineering — Indian Institute of Science, Bangalore, India

BEng, Mechanical Engineering — University of Mumbai, Mumbai, India

Technical Skills

Languages/Data — Python, pandas, Polars, NumPy, SciPy

ML/Optimization — PyTorch, scikit-learn, TabPFN, Chronos, PEFT, HMMs, NSGA-II, Bayesian optimization

LLM/Agents — DSPy, GEPA, LangChain, MCP, multimodal RAG, Qdrant, Groq, Fireworks AI

Cloud/Infrastructure — AWS (Lambda, DynamoDB, S3, ECR, EventBridge, CloudFormation), Docker, Modal, GitHub Actions

Web/Broker APIs — FastAPI, React, Next.js, REST, Alpaca, OANDA, TradeStation